

Project Checkpoint 1

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(A)

1. AI-Driven Fraud as an Emerging Cyber Risk
 - a. Source: Harvard Dataverse
 - b. Course Topic Alignment:
 - i. Clustering (grouping incident by behavior or impact)
 - ii. Frequent itemsets / association rules
 - c. Beyond Course techniques:
 - i. Temporal Trend Analysis
 - ii. Network modeling of country-platform-scam relationships
 - d. Dataset Size and Structure:
 - i. 167 rows
 - ii. Each row is a fraud incident
 - iii. Structured tabular dataset
 - e. Data Types:
 - i. Date (datetime)
 - ii. Text (description)
 - iii. Categorical (scam type, victim type)
 - iv. Numeric (financial loss, number of victims)
 - f. Target Variables: None
 - g. Licensing: CC0

Tulga, Ahmet Yiğitalp, 2026, "AI-Driven Fraud as an Emerging Cyber Risk", <https://doi.org/10.7910/DVN/YBB314>, Harvard Dataverse, V1

2. CIC-DGG 2025

- a. Source: UNB CIC
- b. Course Topic Alignment:
 - i. Graph mining
 - ii. Graph embedding
 - iii. Clustering
 - iv. Anomaly detection
- c. Beyond Course techniques:
 - i. GNNs
 - ii. Graph Representation Learning (DeepWalk / Node2Vec)
- d. Dataset Size and structure:
 - i. 2,659 graph instances
 - ii. Each instance is a separate Control Flow Graph
 - iii. Metadata available via `cfg_map.csv`
- e. Data types:
 - i. Numeric (`number_nodes`, `number_edges`)
 - ii. Categorical (dataset source)
- f. Target variables: label
- g. Licensing: Open license for researchers

H. Shokouhinejad, G. Higgins, R. Razavi-Far, H. Mohammadian, A. Ghorbani. "[On the Consistency of GNN Explanations for Malware Detection](#)," Information Sciences, Dec 2025.

3. LIAR Dataset

- a. Source: PolitiFact fact-checking archive
- b. Course Topic Alignment:
 - i. Clustering
 - ii. Text mining
 - iii. Anomaly detection
- c. Beyond Course techniques:
 - i. Transformer-based embeddings
 - ii. Topic modeling
- d. Dataset size and structure:
 - i. 12,791 statements
 - ii. 16 structured columns
- e. Data types:
 - i. Categorical (label, party, subject)
 - ii. Text (statement)
 - iii. Numeric (truth counts, statement length)
- f. Target variables: label (ordinal)
- g. Licensing: Public academic dataset.

Wang, William. “Liar, Liar Pants on Fire”: A New Benchmark Dataset for Fake News detection”. ArXiv. <https://doi.org/10.48550/arXiv.1705.00648>

(B)

Dimension	AI-Driven Fraud	CIC-DGG 2025 (Graph)	LIAR Dataset
Supported Data Mining Tasks (Course)	Clustering (incident grouping), Frequent Itemsets (co-occurring scam traits)	Graph Mining (centrality, components), Clustering, Anomaly Detection	Text Mining, Clustering, Anomaly Detection
Supported External Techniques	Temporal trend modeling, Incident network modeling	GNNs, Graph Representation Learning	Transformer embeddings (SBERT/BERT),

Dimension	AI-Driven Fraud	CIC-DGG 2025 (Graph)	LIAR Dataset
	(country-platform-scam graphs)	(Node2Vec, DeepWalk), Graph-level embeddings	Topic modeling (LDA/BERTopic), Ordinal modeling
Data Quality Issues	Small dataset (167 rows), possible underreporting bias, inconsistent text descriptions	Very large graph files (50MB+), dependency on static analysis tools (angr), potential graph sparsity	Noisy metadata (state, party), class imbalance (pants-fire small), speaker dominance bias
Algorithmic Feasibility	Easily manageable on laptop; low computational cost	Computationally heavy; graph loading expensive; may require sampling strategy	Fully feasible on laptop; scalable TF-IDF and embedding approaches
Bias Considerations	Reporting bias (only detected/known fraud cases), regional reporting imbalance	Malware datasets may overrepresent certain malware families; benign samples from specific sources	Editorial selection bias; political imbalance; speaker frequency bias
Ethical Considerations	Sensitive financial data context; must avoid overgeneralization	Malware classification research has defensive applications; low direct harm risk	Political labeling risks misuse; fairness and partisan bias concerns

Dimension	AI-Driven Fraud	CIC-DGG 2025 (Graph)	LIAR Dataset
Scale & Complexity	Small, structured tabular	Medium-size dataset but heavy graph complexity	Medium-size structured + unstructured text
Technical Depth Potential	Moderate	High	Moderate–High (if using advanced NLP)
Risk of Overfitting / Data Leakage	High (small sample size)	Medium (graph structural leakage possible)	High (speaker memorization risk)

(C)

The selected dataset for this project is the LIAR dataset, derived from the PolitiFact fact-checking archive. This dataset provides strong alignment with multiple course topics while remaining computationally feasible for experimentation on a personal machine. It directly supports text mining techniques such as TF-IDF feature extraction, supervised classification, clustering, and dimensionality reduction. It also enables anomaly detection approaches by treating extreme categories such as “pants-fire” as potential outliers. Additionally, the dataset naturally supports Locality-Sensitive Hashing for approximate similarity search and near-duplicate claim detection. Beyond course content, the LIAR dataset allows exploration of transformer-based embeddings, topic modeling methods such as LDA or BERTopic, and ordinal classification strategies that account for the inherent ordering of truthfulness labels.

Compared to graph-based malware datasets that require heavy computational resources and specialized tooling, the LIAR dataset offers a strong balance between methodological depth and practical feasibility. While it does present trade-offs, including selection bias, class imbalance, and speaker concentration effects, these

limitations themselves provide meaningful opportunities for analysis, particularly in evaluating model generalization and fairness.

(D)

EDA is found in the ‘.ipynb’ file.

(E)

The exploratory data analysis revealed several important structural patterns in the LIAR dataset. First, truth labels are moderately imbalanced, with “pants-fire” representing the smallest category. Second, statement length distributions are largely similar across truth categories, suggesting that surface-level textual features such as length alone are insufficient for distinguishing truthfulness. Third, clustering using TF-IDF features resulted in an Adjusted Rand Index near zero, indicating that unsupervised lexical similarity does not naturally recover fact-check labels. This suggests that truthfulness depends on deeper semantic and contextual information rather than simple word co-occurrence. Additionally, topic analysis showed that label distributions vary significantly by subject, implying that misinformation may be topic-dependent. Speaker analysis also revealed concentration effects, where a small number of public figures account for a large proportion of statements, raising concerns about potential speaker memorization in supervised models.

These observations motivate several research questions for further investigation:

- Does the use of transformer-based embeddings significantly improve classification performance compared to TF-IDF representations?
- Can topic modeling uncover latent structure that clustering methods failed to capture?
- How does model performance change under speaker-disjoint evaluation compared to random train-test splits?
- Does treating truthfulness as an ordinal problem improve predictive performance over flat multi-class classification?

- Are certain topics or political affiliations associated with systematically different model error rates?

(F)

GitHub Repo: [Here](#)